

Autonomous Technology Researcher Agent

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Project Number:AAI-29

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Academic Year:2025-26

Problem Statement & Objectives

1. Problem Statement

In today's digital world, users rely heavily on search engines to gather information. However, the information available is often scattered, unstructured, and time-consuming to analyse.

Users need to visit multiple sources, read lengthy content, and manually summarize the information.

There is a need for an intelligent system that can automatically research, analyse, and present information in a clear and structured format.

2. Project Objectives

- To generate structured and concise responses from user queries
- To automate the process of information gathering and summarization
- To implement a multi-agent system for better task handling
- To build a user-friendly interface for interaction

3. Scope of the Project

The scope of this project includes:

- Accepting user queries through a web interface
- Processing queries using AI agents
- Generating structured research responses
- Providing real-time interaction with users

The project currently focuses on technology-related queries but can be extended to other domains such as healthcare, finance, and education.

Proposed Solution

1. Key Features

The system offers the following key features:

- AI-powered research generation
- Multi-agent architecture (Research Scout + Industry Analyst + Knowledge Curator)
- Structured and readable output
- Interactive frontend interface

- Real-time response handling

2. Overall Architecture / Workflow

The system follows a client-server architecture with AI agent integration.

Workflow:

User → Frontend → Backend API → AI Agents → LLM → Response → Frontend

Explanation:

The user enters a query in the frontend.

The frontend sends the request to the backend API.

The backend creates tasks and assigns them to AI agents.

The agents process the query using a Large Language Model and generate a structured response.

Finally, the response is sent back to the frontend and displayed to the user.

3. Tools & Technologies Used

The project uses the following technologies:

♦ Frontend:

- React.js + Vite
- CSS

♦ Backend:

- FastAPI

♦ AI & Processing:

- CrewAI (Agent orchestration)
- LangChain (LLM integration)

♦ LLM Provider:

- Groq

Other Tools:

- Python
- JavaScript
- REST APIs

3. Key Outcomes

The project successfully achieved the following outcomes:

Functional Outcomes:

- Successfully built a full-stack AI application
- Implemented multi-agent system (Research + Writer)
- Generated structured and meaningful responses
- Integrated frontend with backend API

Performance Outcomes:

- System responds dynamically to user queries
- Supports real-time interaction
- Handles multiple queries independently

AI Outcomes:

- Demonstrates agent-based AI workflow
- Produces human-like responses
- Automates research and summarization

Overall Outcome:

The system effectively reduces the effort required to gather and summarize information by automating the research process using AI agents.

4. Results & Output

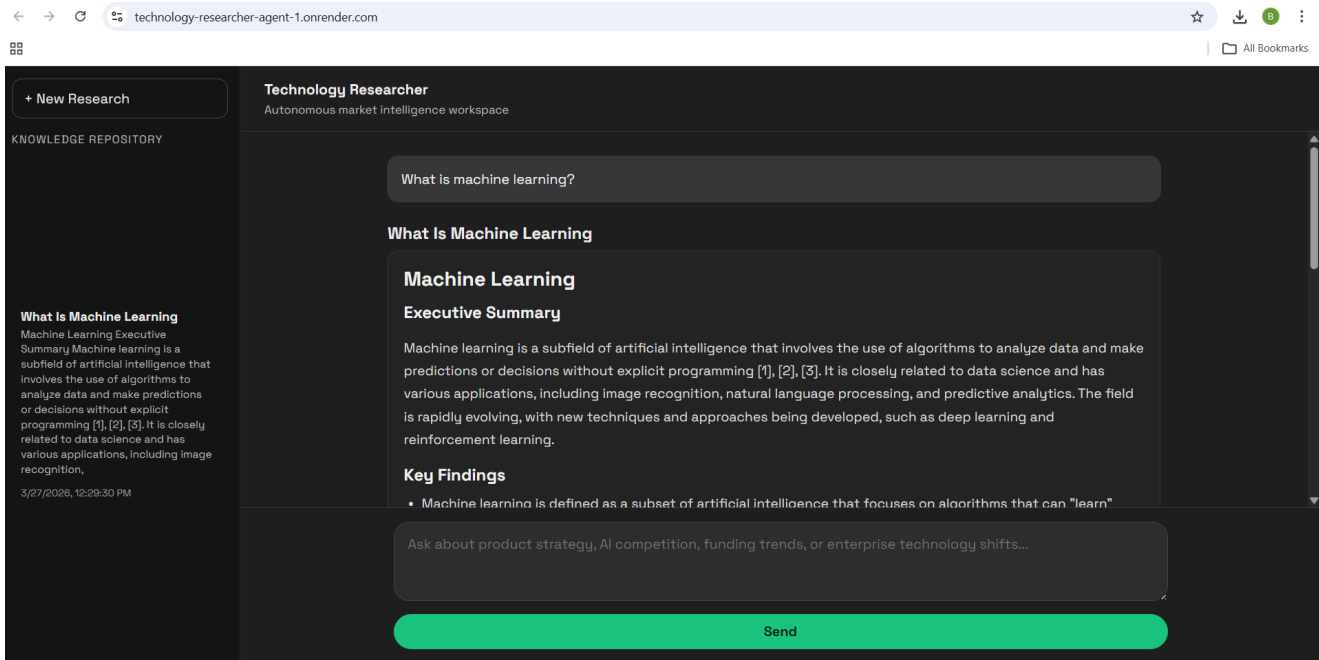


Fig. 4.1 Technology Research Agent Dashboard

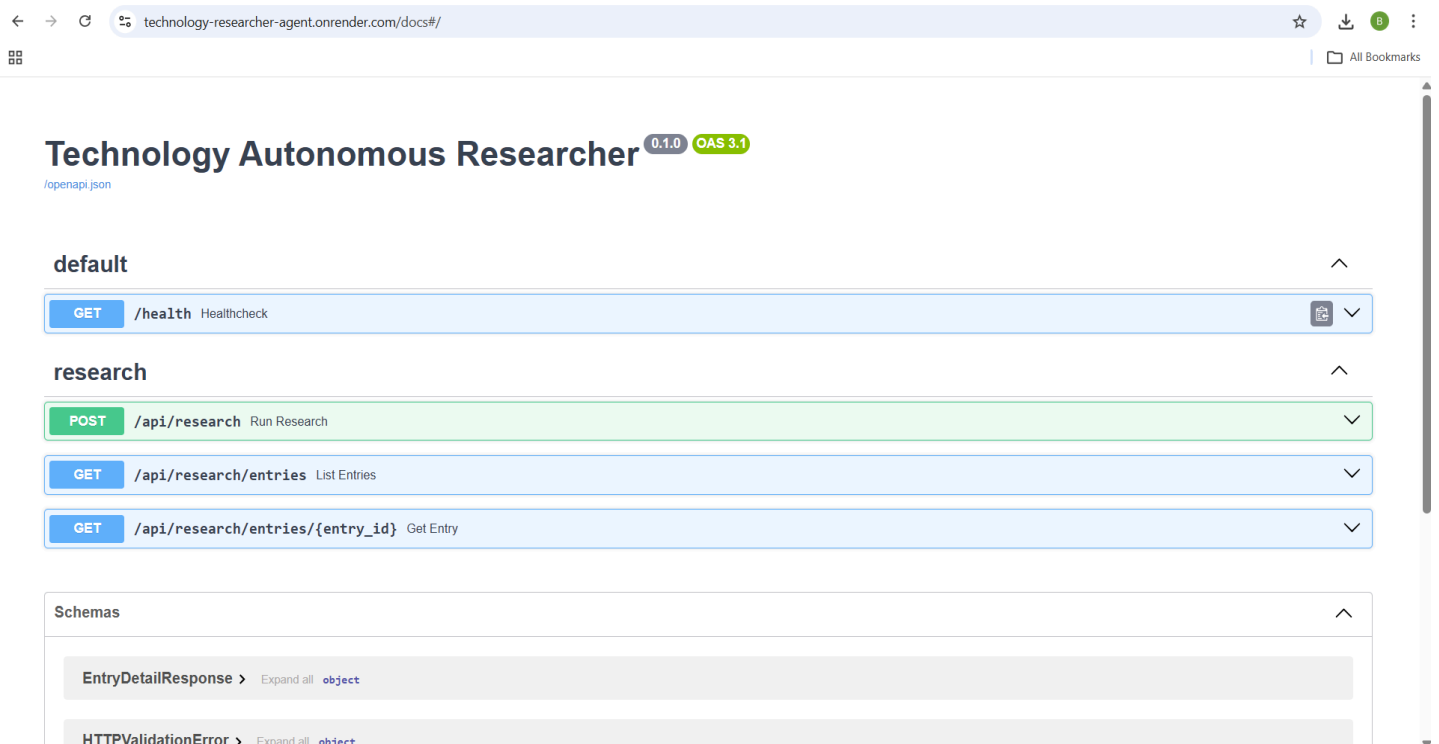


Fig. 4.2 Technology Research Agent Backend Service

+ New Research

KNOWLEDGE REPOSITORY

What Is Ai

What is AI Executive Summary
Artificial Intelligence (AI) refers to a set of technologies that enable computers to learn, reason, and perform advanced tasks that typically require human intelligence [1]. The field of AI encompasses various types, including Weak AI, Strong AI, Artificial General Intelligence, and Arti

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What Is Machine Learning

Machine Learning Executive Summary Machine learning is a subfield of artificial intelligence that involves the use of algorithms to analyze data and make predictions or decisions without explicit programming [1], [2], [3]. It is closely related to data science and has various applications, including image recognition,

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Fig. 4.3 Knowledge Repository

Conclusion

The Technology Research Agent project successfully demonstrates the integration of modern web technologies with artificial intelligence to automate the process of information gathering and summarization.

The system allows users to input queries and receive structured, meaningful responses generated through a multi-agent architecture. By utilizing CrewAI for agent orchestration and LangChain for LLM integration, the project effectively showcases how intelligent agents can collaborate to perform complex tasks.

The frontend built using Next.js and the backend developed with FastAPI provide a seamless user experience, enabling smooth interaction between users and the AI system.

Future Scope & Enhancements

Future Scope

The AI Technology Research Agent has significant potential for future expansion and real-world applications.

Possible Extensions:

- **Multi-domain Support**

Extend the system beyond technology to domains like healthcare, finance, education, and law.

- **Conversational AI (Chat System)**

Transform the system into a chatbot with continuous interaction instead of single-query responses.

- **Integration with External Data Sources**

Connect with live APIs, databases, and search engines for real-time and updated information.

Enhancements

“The following enhancements can improve performance, usability, and scalability of the system:”

1. Performance Improvements

- Implement **streaming responses** for faster output
- Use optimized LLMs (like Groq)
- Reduce latency in agent processing

2. Memory & Context Awareness

- Add chat history
- Maintain session continuity

3. Database Integration

- Store user queries and responses
- Enable history tracking
- Use databases like MongoDB or PostgreSQL

4. UI/UX Enhancements

- Improve frontend design (dark mode, animations)
- Add chat-style interface

5. Caching Mechanism

- Cache repeated queries
- Reduce API calls and cost
- Improve response time

6. Security Enhancements

- Add authentication (login/signup)
- Secure API endpoints

